ORGANIC NAMED REACTIONS

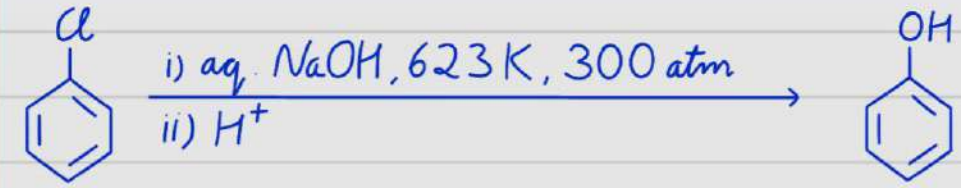
ALKYL AND ARYL HALIDE

Hans diya kar, raat ko ayegi baarat cycle par

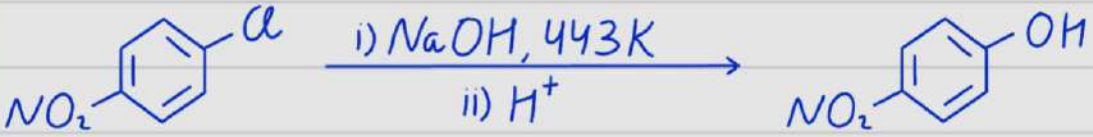
1. **Borodine Hunsdiecker Reaction :**

$$RCOOAg + Br_2 \xrightarrow{CCl_4} RBr + AgBr + CO_2$$

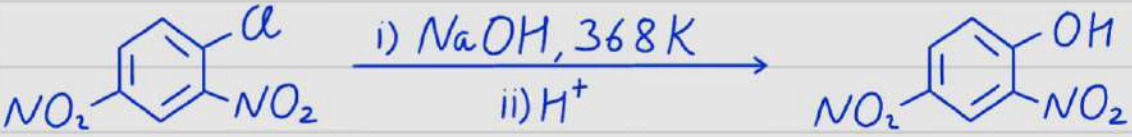
2. **Dow's Process :**



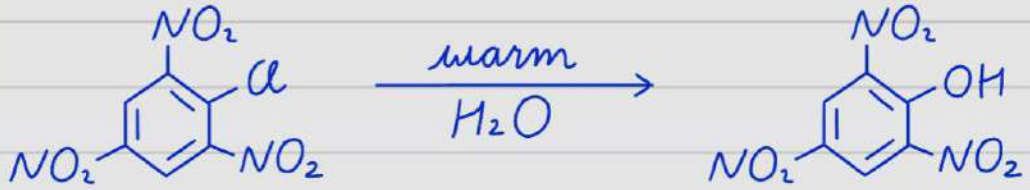
i)



ii)



iii)



3. **Finkelstein Reaction :** Finkel गाड़ी A.C. में

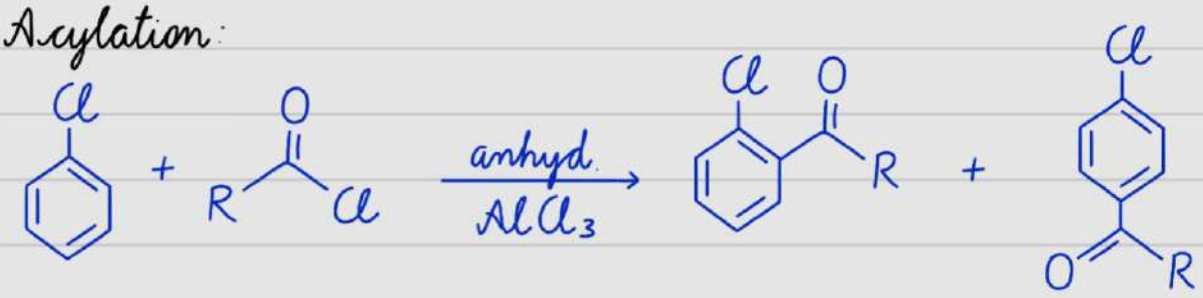
$$RX + NaI \xrightarrow{\text{Acetone}} RI + NaX \quad \text{where } X = Cl, Br$$

4. **Grignard's Reagent :**

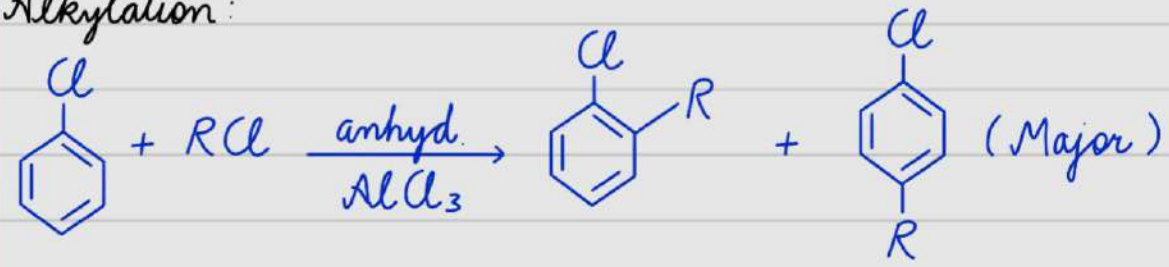
$$RX + Mg \xrightarrow{\text{dry ether}} RMgX$$

5. **Friedal - Crafts Reaction :**

i) Acylation:



ii) Alkylation:



6. **Swarts Reaction:**

$$RX + AgF / Hg_2F_2 / SbF_3 \longrightarrow RF \quad \text{where } X = Cl, Br$$

7. **Williamson's Synthesis:** WS bahut rota tha (Rona Rxn)

$$RX + R'ONa \xrightarrow{\text{dry ether}} ROR' + NaX$$

8. **Wurtz's Reaction:** WhatsApp FaceBook; Na घटी Na

$$2RX + 2Na \xrightarrow{\text{dry ether}} R\text{-}R + 2NaX$$

9. **Fittig Reaction:**



10. **Wurtz-Fittig Reaction:**



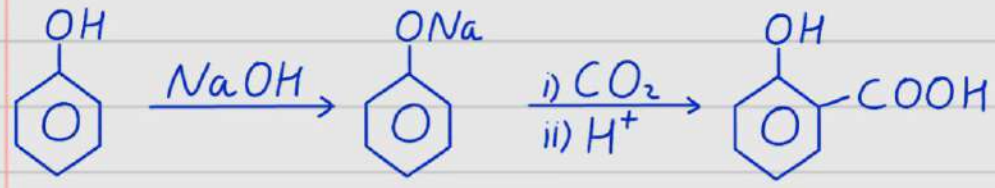
ALCOHOLS, PHENOLS AND ETHERS

1. **Luca's Test (dry $HCl + ZnCl_2$):**

$ROH \xrightarrow{\text{Luca's Reag}} RCl \downarrow$ (White ppt)

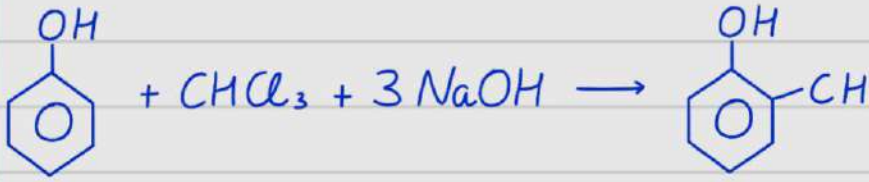
3° Alcohol $\longrightarrow$ White ppt : Immediately
2° " $\longrightarrow$ " : After 5-10 mins
1° " $\longrightarrow$ No white ppt at room temp.

2. **Kolbe's Reaction:**



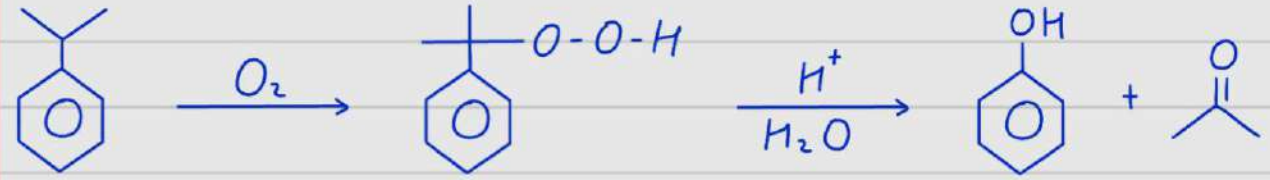
Salicylic Acid

3. **Reimer - Tiemann Reaction:** R-T mei phenol ko base mei dissolve kar chloroform sungha diya



Salicylaldehyde

4. **Cumene Process:** कुमीन ने O_2 लिया, फिर पानी

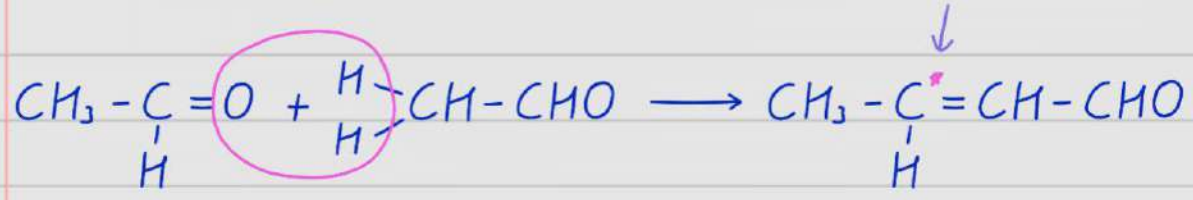


7. Aldol Condensation:

Aldehydes & ketones having at least one α-H undergo a rxn in the presence of dil alkali as catalyst to form β-hydroxy aldehydes (aldol) or β-hydroxy ketones (ketols).

$$2CH_3CHO + \text{dil } NaOH \longrightarrow \text{Aldol} \xrightarrow{\Delta} \text{AC Product}$$

Trick:



For aldol, add OH to C* & H to another C in ACP.

Aldol: $CH_3-\overset{OH}{\overset{|}{C}}H-CH_2-CHO$

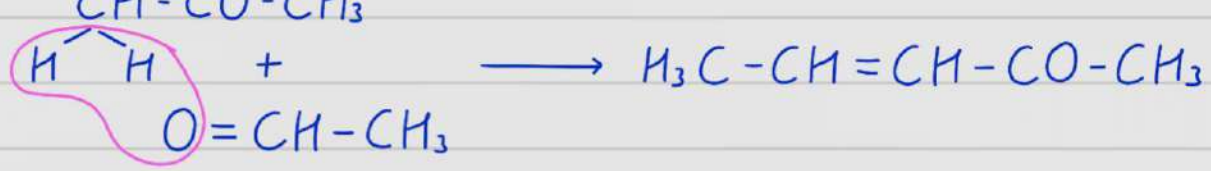
8. Cross aldol condensation:

If both have α-H, then products → 4 (2 self, 2 cross)

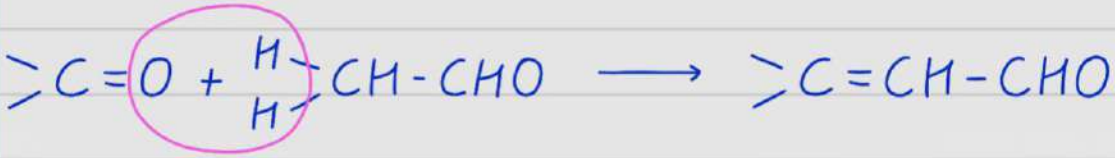
If one has α-H, then products → 2 (1 self, 1 cross)

Eg: $CH_3CHO + CH_3COCH_3 \xrightarrow[ii)\ \Delta]{i)\ NaOH}$

Cross: i)



ii)



9. Cannizaro Rxn:

Aldehydes with no α-H undergo self oxidation & reduction (disproportionation) rxn.

ALDEHYDES, KETONS AND CARBOXYLIC ACIDS

1. **Rosenmund's Reduction:**

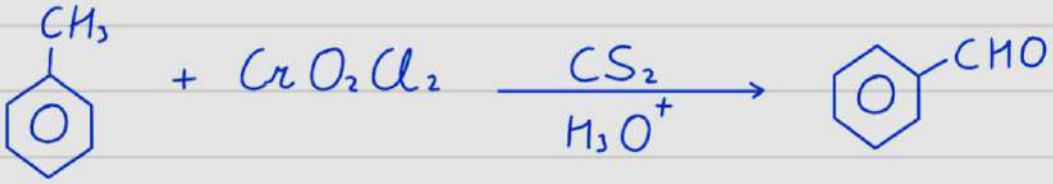
$$R-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow[(Quinolene)]{Pd-BaSO_4, H_2} R-\overset{\overset{O}{\|}}{C}-H$$

where $Pd-BaSO_4$ is 'Lindlar's catalyst' poisened by S.

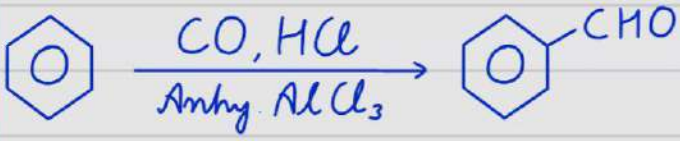
2. **Stephan's Reduction:**

$$R-C\equiv N \xrightarrow[HCl]{SnCl_2} \underset{\text{imine}}{R-CH=NH} \xrightarrow{H_3O^+} RCHO$$

3. **Etard Reaction:**



4. **Gatterman-Koch Reaction:**

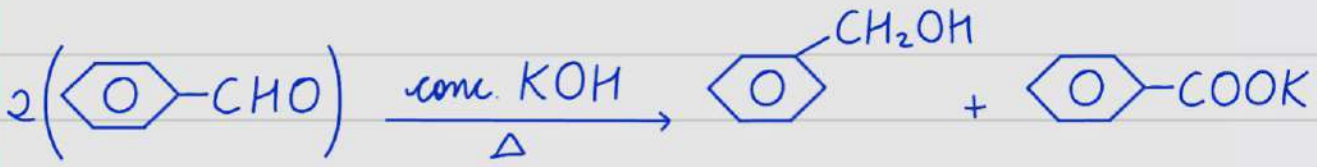


5. **Clemensen Reduction:**

$$>C=O \xrightarrow[HCl]{Zn-Hg} >CH_2 + H_2O$$

6. **Wolff-Kishner Reduction:**

$$>C=O \xrightarrow[-H_2O]{NH_2NH_2} >C=NNH_2 \xrightarrow[KOH,\ \Delta]{\text{ethylene glycol /}} >CH_2$$



10. HVZ (RCOOH should have α-H):

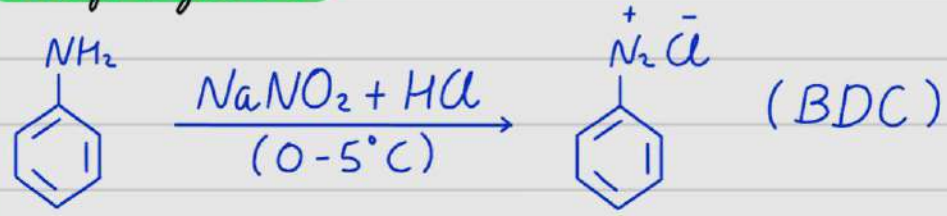
$$RCH_2\text{-}COOH \xrightarrow[ii)\ H_2O]{i)\ Br_2/\text{Red P}} R\text{-}\underset{\displaystyle Br}{\underset{|}{C}}H\text{-}COOH$$

AMINES

1. **Hoffmann Ammonolysis:**

$$RX + NH_3 \longrightarrow RNH_2 \xrightarrow{RX} R_2NH \xrightarrow{RX} R_3N \xrightarrow{RX} (R_4N)X$$

2. **Diazotization:**



3. **Sandmeyer Reaction:**

$$ArN_2^+X^- \xrightarrow[]{\begin{array}{l} Cu_2Cl_2/HCl \\ Cu_2Br_2/HBr \\ CuCN/KCN \end{array}} \begin{array}{l} ArCl \\ ArBr \\ ArCN \end{array}$$

4. **Gattermann Reaction:** Sand mei salt, gutter mei powder

$$ArN_2^+X^- \xrightarrow[]{\begin{array}{l} Cu/HCl \\ Cu/HBr \end{array}} \begin{array}{l} ArCl \\ ArBr \end{array}$$

5. **Balz-Schiemann Reaction:** Bal G hai bwafa

$$ArN_2^+Cl^- + HBF_4 \longrightarrow ArN_2^+BF_4^- \xrightarrow{\Delta} ArF$$